

## Theory and Methodology

# A model for the relative importance of the criteria in the Multiplicative AHP and SMART

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**Abstract**

We define the relative importance of any pair of criteria as the substitution rate between the relative gains and losses of the alternatives when we move along an indifference curve. Under the geometric-mean aggregation rule in the Multiplicative AHP and under the arithmetic-mean aggregation rule in SMART, the relative (not the marginal) substitution rate depends neither on the performance of the alternatives under the remaining criteria nor on the units of performance measurement. Hence, it provides a sound argument for distributed decision-making processes where those who judge the criteria are not the same actors as those who assess the performance of the alternatives. The definition has a plausible basis in the psycho-physical research on the relationship between physical stimuli and sensory responses, which shows that human beings are sensitive, not to marginal but to relative changes of the stimulus intensities.

*Keywords:* Multi-criteria analysis; Pairwise comparisons; Geometric scales; Direct rating; Aggregation; Substitution rates; Power functions

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**1. Introduction**

The question of how to weigh the importance of the criteria in applications of multi-criteria decision analysis is a delicate one. At first sight, criterion weights are relevant in a particular situation only. They depend on the physical or monetary units expressing the performance of the alternatives under each of the criteria separately. They also depend on the manner in which they will be used, that is, on the aggregation rule generating the final scores which express the performance of the alternatives under all criteria simultaneously. Finally, the relative importance of any pair of criteria, possibly expressed by the ratio of the corresponding criterion weights, depends on the performance of the alternatives under the other criteria in the given decision problem.

Usually, however, decision makers seem to ignore these issues. They are prepared to estimate the weights of the criteria or their relative importance, regardless of how the performance of the alternatives has or will be recorded and regardless of the aggregation procedure to be used in the subsequent analysis, so that they seem to supply meaningless information (see Belton, 1986, Vincke, 1989, and Mousseau, 1993, for instance, as well as Borcherding et al., 1994, who discuss several biases in the elicitation of criterion weights). The underlying reason may be that criteria have emotional or social ‘values’ which neither depend on the actual decision problem itself nor on the method of analysis. Decision makers with managerial or administrative responsibilities over a long period of time, in fields such as energy planning, water management, health care, and environ-

mental protection, feel the need to be consistent, that is, to use a coherent set of criteria in all their decision problems, with roughly the same weights or the same relative importance. In fact, decision makers frequently judge the 'values' of the criteria in isolation from a particular decision problem, whereas they have to provide a full justification of their decisions afterwards (Stewart, 1992). In an administrative or managerial hierarchy the situation may be even more complicated because the choice of criteria and the vague verbal formulation of their relative importance is frequently felt to be the prerogative of those who established a committee to evaluate certain alternatives. Obviously, we have a distributed decision process here, because those who judge the relative importance of the criteria and those who judge the alternatives under the respective criteria are not the same decision makers or actors in the process. Hierarchically, they may be one or more levels apart.

The purpose of the present paper is to support such a distributed decision process and to find relaxation conditions which allow the decision makers to ignore the units of performance measurement as well as the performance levels of the alternatives when the importance of the criteria is assessed. This is achieved by a new model for the relative importance of the criteria in two popular methods for multi-criteria decision analysis: the Multiplicative AHP (Lootsma, 1993), a multiplicative variant of the Analytic Hierarchy Process (Saaty, 1980), and SMART (the Simple Multi-Attribute Rating Technique, see Von Winterfeldt and Edwards, 1986). The model, in fact the ratio of the relative gains and losses of the alternatives during a move along an indifference curve, enables the decision makers to judge the relative importance of the criteria in a particular field of problems but in the absence of immediate context, at least under the geometric-mean aggregation rule which is closely related to the power function in the general psycho-physical law for stimulus-response relationships (Stevens, 1957).

The organization of the paper is as follows. In Section 2, we briefly describe the geometric scale for the quantification of verbal comparative judgement in the Multiplicative AHP, as well as the calculation of the impact scores of the alternatives (the partial scores under each of the criteria individually), and

we introduce the weighted geometric mean to aggregate the impact scores into final scores. Section 3 presents some examples to show the critical issues when we work with arithmetic and geometric means; special attention is given to the role of the units of performance measurement and to the compatibility of the reference points on the measurement scales. Section 4 introduces the new definition of the relative importance of the criteria as the substitution rate between the relative gains and losses of the alternatives along an indifference curve. Section 5 is concerned with the choice of scale values which correspond to verbal comparative statements such as 'somewhat more', 'definitely more', 'much more', and 'vastly more' important. In Section 6 we discuss the question of where the results of this paper belong: in the constructivism of the French school of multi-criteria analysis, or in the clinical approach of the American school. Finally, Section 7 sketches the significance of the results in the analysis of a hierarchy of criteria, sub-criteria, sub-sub-criteria, etc. Such hierarchies are widely advocated in the literature on multi-criteria decision analysis, although there was no plausible interpretation of the relative importance of the sub-criteria, etc. The common view has hitherto been that importance seems to be some sort of substance that can arbitrarily be fragmented and added together.

## 2. Pairwise comparisons and the calculation of impact scores

Let us first introduce some notation. We consider a decision problem where the alternatives  $A_j$ ,  $j = 1, \dots, n$ , must be evaluated under the criteria  $C_i$ ,  $i = 1, \dots, m$ , by a group of decision makers  $d = 1, \dots, g$ . In the basic experiment at the first evaluation level of the Multiplicative AHP, two stimuli  $S_j$  and  $S_k$  (two alternatives  $A_j$  and  $A_k$  under a particular criterion) are presented to decision maker  $d$  who is requested to express his/her graded comparative judgement, that is, to express his/her indifference between the two, or his/her weak, definite, strong, or very strong preference for one of them. We assume that the stimuli have unknown subjective values  $V_j$  and  $V_k$  which are the same for all decision makers in the group (otherwise, a compromise would

hardly be possible, whereas the group members are usually supposed to arrive jointly at a common group standpoint). The purpose of the basic experiments and the subsequent analysis is to approximate these values by the calculated impact scores. The verbal comparative judgement, given by decision maker  $d$  and converted into a numerical value  $r_{jkd}$ , is taken to be an estimate of the ratio  $V_j/V_k$ . Hence, since we only have ratio information, we may take the subjective values to be normalized in the sense that they sum to 1 or to 100%.

In the Multiplicative AHP we convert the gradations of the decision makers' comparative judgement into numerical values on a geometric scale which is conveniently characterized by the scale parameter  $\gamma$ . Thus, we set

$$r_{jkd} = \exp(\gamma \delta_{jkd}),$$

where  $\delta_{jkd}$  is an integer-valued index designating the gradations of the decision maker's comparative judgement as follows:

- 8: Very strong preference for  $S_k$  versus  $S_j$ .
- 6: Strong preference for  $S_k$  versus  $S_j$ .
- 4: Definite preference for  $S_k$  versus  $S_j$ .
- 2: Weak preference for  $S_k$  versus  $S_j$ .
- 0: Indifference between  $S_j$  and  $S_k$ .
- + 2: Weak preference for  $S_j$  versus  $S_k$ .
- + 4: Definite preference for  $S_j$  versus  $S_k$ .
- + 6: Strong preference for  $S_j$  versus  $S_k$ .
- + 8: Very strong preference for  $S_j$  versus  $S_k$ .

Intermediate integer values may be assigned to the gradation index in order to express hesitations between two adjacent gradations. A plausible value of the scale parameter  $\gamma$  is given by  $\ln 2$ , which implies that we are working on a geometric scale with progression factor 2 (see Lootsma, 1993).

Next, we approximate the vector  $V$  of subjective stimulus values via the logarithmic least-squares method. Taking the set  $D_{jk}$  to denote the set of decision makers who judged  $S_j$  with respect to  $S_k$  we approximate the vector  $V$  by the normalized vector  $\bar{v}$  which minimizes the function

$$\sum_{j < k} \sum_{d \in D_{jk}} (\ln r_{jkd} - \ln v_j + \ln v_k)^2. \quad (1)$$

Introducing the quantities

$$q_{jkd} = \ln r_{jkd} = \gamma \delta_{jkd}$$

and

$$w_j = \ln v_j,$$

we can rewrite (1) as

$$\sum_{j < k} \sum_{d \in D_{jk}} (q_{jkd} - w_j + w_k)^2. \quad (2)$$

Thus, we have a quadratic function in the variables  $w_j$ ,  $j = 1, \dots, n$ . Setting the first-order derivatives of the least-squares function (2) to zero, we obtain the so-called normal equations associated with the problem of minimizing (2). Let the vector  $w^*$  with components  $w_j^*$ ,  $j = 1, \dots, n$ , denote a solution to these equations, then the vector  $v^*$  with components

$$v_j^* = \exp(w_j^*), \quad j = 1, \dots, n,$$

yields a solution to the original logarithmic least-squares problem of minimizing (1). Since there is an additive degree of freedom in the solution  $w^*$  of (2), there is a multiplicative degree of freedom in  $v^*$  which can be used to find the normalized solution  $\bar{v}$  (for detailed calculations, see Lootsma, 1993).

This procedure is carried out  $m$  times, under each of the criteria separately. Let us now consider two alternatives  $A_j$  and  $A_k$  with their calculated, normalized weights (the so-called impact scores or partial scores)  $\bar{v}_{ij}$  and  $\bar{v}_{ik}$  under criterion  $C_i$ . For each  $i$  the preference ratio

$$\bar{v}_{ij}/\bar{v}_{ik}$$

expressing the relative preference for  $A_j$  with respect to  $A_k$  under  $C_i$  is unique. Since we are dealing with ratios it is natural to model the global (aggregate) preference for  $A_j$  with respect to  $A_k$  by a geometric mean. This will be further explained in Section 3. Taking  $\bar{c}_i$  to stand for the calculated weight of criterion  $C_i$  and assuming that these weights are normalized we represent the global preference for  $A_j$  with respect to  $A_k$  by the weighted geometric mean

$$\prod_{i=1}^m (\bar{v}_{ij}/\bar{v}_{ik})^{\bar{c}_i}. \quad (3)$$

In general, we may set the final score of the alternative  $A_j$  to

$$\bar{f}_j = \alpha \prod_{i=1}^m (\beta_i \bar{v}_{ij})^{\bar{c}_i}, \quad (4)$$

where  $\alpha$  is a general multiplicative degree of freedom and  $\beta_i$  the multiplicative degree of freedom in the impact scores under criterion  $C_i$ . Nevertheless, the ratio of any two final scores is unique (see formula (3)). Usually, normalized impact scores ( $\beta_i = 1$ ) and normalized final scores (by a proper choice of  $\alpha$ ) are presented to the decision maker. Normalization is a cosmetic operation, however, which only makes the results easier to read.

### 3. Intermezzo on arithmetic and geometric means

Aggregation via weighted geometric means is still not widely used in applications of the AHP, despite its advantages. Barzilai et al. (1987) proposed such a procedure on axiomatic grounds (see also Barzilai, 1992), and Lootsma (1993) demonstrated that some notorious occurrences of rank reversal can be avoided by this type of aggregation. In this section we sketch via a number of examples why the weighted geometric mean seems to be an appropriate model for global (aggregate) preference when we are dealing with ratio information.

Let us recall that, in order to express the position of a point on a one-dimensional axis by a real number, the so-called coordinate, we need two anchor points or, equivalently, one anchor point and a unit of measurement, so that there are in general two degrees of freedom in the coordinates. Ratios of differences of interval lengths on that axis do not depend on the anchor points. Usually referred to as dimensionless quantities, they are uniquely expressed as ratios of differences of the corresponding coordinates. If one of the anchor points is a natural zero on the axis under consideration (zero length, zero weight, etc.), there is one degree of freedom only, the unit of measurement, so that ratios of distances from zero, uniquely expressed as ratios of the corresponding coordinates, are also dimensionless quantities.

First, we consider in general a number of quantities  $x_i$ ,  $i = 1, \dots, m$ , and we concern ourselves with the question of how to find a quantity  $\bar{x}$  such that the deviations  $d(x_i, \bar{x})$  between  $x_i$  and  $\bar{x}$ , all taken together, cancel each other. The result depends on how the deviations are measured. The arithmetic mean is found when the deviations are expressed as

differences. Setting the sum of the deviations to zero, we obtain

$$\sum_{i=1}^m (x_i - \bar{x}) = 0,$$

whence

$$\bar{x} = \frac{1}{m} \sum_{i=1}^m x_i.$$

The geometric mean is found when the deviations are expressed as ratios. Setting the product of the deviations to one, we obtain

$$\prod_{i=1}^m (x_i / \bar{x}) = 1,$$

whence

$$\bar{x} = \sqrt[m]{\prod_{i=1}^m x_i}.$$

Usually, these operations make sense for a decision maker if all deviations are expressed in the same physical or monetary units (arithmetic means), or if they are all dimensionless quantities (geometric means). Let us now illustrate the role of the units of measurement via a number of examples.

1. Arithmetic means. Consider two locations  $L_j$  and  $L_k$ , and the respective temperatures  $t_{ij}$  and  $t_{ik}$  at these locations measured at noon on the days  $i = 1, \dots, m$ . The observations were made, on some days by a European who recorded the temperatures in °C, and on the remaining days by an American who noted the temperatures in °F. The average difference of the noon temperatures at these locations is given by the arithmetic mean.

$$\frac{1}{m} \sum_{i=1}^m (t_{ij} - t_{ik}),$$

a quantity which only makes sense for a decision maker if some temperatures are rescaled so that they are all expressed in the same units.

2. Geometric means. Consider two programmed algorithms  $A_j$  and  $A_k$ , and the respective execution times  $t_{ij}$  and  $t_{ik}$  to solve the test problems  $i = 1, \dots, m$ . Some of these execution times are expressed in seconds (for the small test problems), others in minutes or hours (for the large test problems). It does not make sense for a decision maker to compare the efficiency of the algorithms on the basis

of the arithmetic mean of the  $t_{ij} - t_{ik}$  (after a conversion into the same time units) because in such a procedure the large problems dominate the small ones. The average execution-time ratio given by the geometric mean

$$\sqrt[m]{\prod_{i=1}^m (t_{ij}/t_{ik})}$$

makes sense, however. Under the condition that all test problems have the same importance, its inverse expresses globally the relative efficiency of the programmed algorithms (since smaller execution times designate a better performance). Its numerical value does not depend on the time units since all ratios in it are dimensionless quantities.

3. Geometric means on a new common dimension. Consider again two programmed algorithms  $A_j$  and  $A_k$ , the respective execution times  $t_{ij}$  and  $t_{ik}$  to solve the test problems  $i = 1, \dots, m$ , as well as the associated space requirements  $s_{ij}$  and  $s_{ik}$ , some of them expressed in kilobytes, others in megabytes. The geometric mean

$$\sqrt[2m]{\prod_{i=1}^m (t_{ij}/t_{ik})(s_{ij}/s_{ik})}$$

does not immediately make sense for a decision maker. Although the ratios in the above expression are dimensionless quantities, some of them are ratios on the dimension of time (seconds, minutes, hours), others are ratios on the dimension of space (kilobytes and megabytes). The geometric mean makes sense, however, if we make a transition to a new common dimension such as quality or desirability. If ‘ $p$  times faster’ means ‘ $p$  times more desirable’, for instance, if ratios of space requirements are interpreted in the same way, and if all ratios have the same importance, then the inverse of the above geometric mean may be used to stand for the desirability ratio of the programmed algorithms.

4. Compatibility of reference points. Suppose that a decision maker considers the locations  $L_j$  and  $L_k$  again, and that he asks how much warmer the first location is compared with the second location. Ratios of the recorded temperatures do not make sense, but ratios of differences with respect to reference temperatures on the respective scales can provide meaningful information. The reference temperatures should be compatible, however. The decision maker cannot simply use the anchor points of the scales

themselves. The freezing points  $0^\circ\text{C}$  and  $32^\circ\text{F}$  of water constitute a pair of compatible reference temperatures, for instance, as well as the lower limits  $-18^\circ\text{C}$  and  $0^\circ\text{F}$  where outdoors human life begins to be feasible until the upper limits  $38^\circ\text{C}$  and  $100^\circ\text{F}$  are attained; the choice of a pair of reference temperatures has to be made by the decision maker. Taking the symbol  $t_i^{\min}$  to represent the reference temperature at the  $i$ -th day, we can use the geometric mean

$$\sqrt[m]{\prod_{i=1}^m \left( \frac{t_{ij} - t_i^{\min}}{t_{ik} - t_i^{\min}} \right)}$$

to express the relative warmth of the first with respect to the second location. Note that we do not have to rescale the recorded temperatures. If they are all expressed in the same units, however, we only have one reference temperature  $t^{\min}$ .

With these considerations in mind, we introduced the dimension of desirability (in an abstract sense) as the new, common dimension for an aggregation procedure (Lootsma, 1993). Taking  $D_{ij}$  and  $D_{ik}$  to represent the desirability of the alternatives  $A_j$  and  $A_k$  under criterion  $C_i$ , and  $D_{\max}$  as the reference point (the common target), we modelled the global preference for  $A_j$  with respect to  $A_k$  as the weighted geometric mean

$$\sqrt[m]{\prod_{i=1}^m \left( \frac{D_{\max} - D_{ik}}{D_{\max} - D_{ij}} \right)^{c_i}},$$

with normalized criterion weights  $c_i$ . We suppose that the weighted geometric mean (3) is an approximation to it. The assumption that there is a common maximum desirability  $D_{\max}$  under all criteria implies that the decision maker’s preferential feelings have a common target and that he/she is not swept away by his/her feelings under some of the criteria.

#### 4. A model for the relative importance of the criteria

Returning to formula (4) where we interrupted the treatment of our subject we consider the function

$$f(v_1, \dots, v_m) = \alpha \prod_{i=1}^m (\beta_i v_i)^{c_i}, \quad (5)$$

underlying the aggregation procedure described at the end of Section 2. The first-order partial derivatives of (5) are given by

$$\frac{\partial f}{\partial v_i} = \frac{c_i}{v_i} f,$$

whence

$$\left( \frac{1}{v_{i_1}} \frac{\partial f}{\partial v_{i_2}} \right) / \left( \frac{1}{v_{i_2}} \frac{\partial f}{\partial v_{i_1}} \right) = \frac{c_{i_2}}{c_{i_1}}, \quad (6)$$

for arbitrary  $i_1$  and  $i_2$ , and regardless of the degrees of freedom  $\alpha, \beta_1, \dots, \beta_m$ . We are now in the position to study the behaviour of  $f$  as a function of  $v_{i_1}$  and  $v_{i_2}$  only, with fixed performance levels  $v_i^*$ ,  $i \neq i_1, i_2$ . So, we consider a contour or indifference curve of  $f$  in the plane  $P^*$  parallel to the  $(v_{i_1}, v_{i_2})$ -coordinate space and through the arbitrary point  $v^* = (v_1^*, \dots, v_m^*)$ . In a first-order approximation, a move from the point  $(v_{i_1}^*, v_{i_2}^*)$  in  $P^*$  towards points (alternatives) in  $P^*$  for which the decision maker is indifferent proceeds in a direction which is orthogonal to the gradient of  $f$  at  $(v_{i_1}^*, v_{i_2}^*)$ , that is, in the direction

$$\left( \frac{\partial f}{\partial v_{i_2}}, -\frac{\partial f}{\partial v_{i_1}} \right).$$

This is shown in Fig. 1. Traditionally (see Keeney and Raiffa, 1976), the ratio

$$\left( \frac{\partial f}{\partial v_{i_2}} \right) / \left( \frac{\partial f}{\partial v_{i_1}} \right) \quad (7)$$

has been defined as the marginal trade-off or the marginal substitution rate between the two criteria under consideration, but with the function (5) this ratio is not constant over the entire plane  $P^*$ . The substitution rate (7) between marginal gains and losses is not an appropriate model for the perceived substitution of properties either. We introduce the relative substitution rate here (the left-hand side of (6)), which is based upon the observation that human beings generally perceive relative gains and losses, that is, gains and losses in relation to the levels from which the move starts. Thus, when a small step is made along the indifference curve, the relative gain (or loss) in the  $v_{i_1}$ -direction is proportional to

$$\frac{1}{v_{i_1}} \frac{\partial f}{\partial v_{i_2}},$$

$$\text{Geometric-mean aggregation rule } f(v) = \alpha \prod_{i=1}^m (\beta_i v_i)^{c_i}.$$

$$\text{Indifference curve in } (v_{i_1}, v_{i_2})\text{-space: } f(v_{i_1}, v_{i_2}) = K.$$

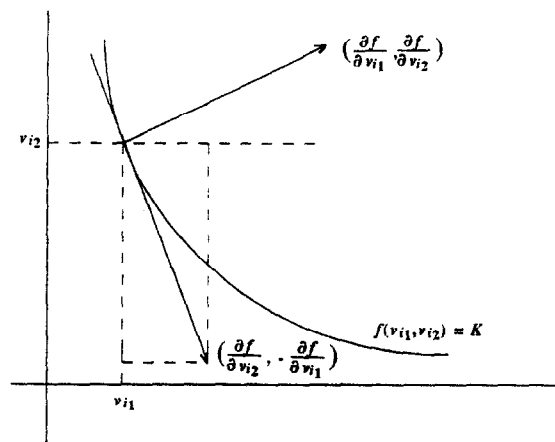


Fig. 1. In a first-order approximation, a move along an indifference curve proceeds in a direction which is orthogonal to the gradient where the move starts. The marginal substitution rate is the ratio of the components of that direction. The relative substitution rate is the ratio of the components divided by the coordinates of the point of departure.

and the corresponding relative loss (or gain) in the  $v_{i_2}$ -direction is proportional to

$$\frac{1}{v_{i_2}} \frac{\partial f}{\partial v_{i_1}}.$$

Under the geometric-mean aggregation rule (which is based upon the function (5)), the substitution rate between relative gains and losses (the left-hand side of (6)) is a constant over the entire plane  $P^*$ , and it does not depend on the performance levels  $v_i^*$ ,  $i \neq i_1, i_2$ . Note that these gains and losses are increments and decrements in subjective values, and that the Multiplicative AHP accordingly aggregates the preferential feelings of the decision makers! Since the left-hand side of (6) is a dimensionless quantity which does not depend on the units of measurement either, decision makers and experts in multi-criteria analysis can meaningfully refer to it as a mathematical model for the relative importance of the criteria in a particular field, in isolation from the particular decision problems in that field. In fact, formula (6) presents an inverse proportionality: if  $c_{i_2} > c_{i_1}$  and if we move along an indifference curve, then a larger

relative step in the  $v_{i_1}$ -direction is compensated by a smaller relative step in the  $v_{i_2}$ -direction. This is just what we would expect when the criteria have different weights in the mind of the decision maker.

The observation that human beings are sensitive, not to marginal but to relative changes, is the result of a century of psycho-physical research on the relationship between the intensity of physical stimuli on the one hand and the intensity of the sensory responses on the other. Weber's law of 1834 states that the just-noticeable difference  $\Delta s$  of two stimulus intensities is proportional to the actual stimulus level  $s$  itself. In Fechner's law of 1860, the sensory response  $\Delta\Psi$  to a just-noticeable difference  $\Delta s$  is supposed to be constant, which implies that  $\Delta\Psi$  would be proportional to  $\Delta s/s$ . Integration yields a logarithmic relationship between  $\Psi$  and  $s$ . Additional experience has finally shown that Fechner's law does in general not hold. Brentano suggested in 1874 that the sensory response  $\Delta\Psi$  to a just-noticeable difference  $\Delta s$  might be proportional to the response level  $\Psi$  itself, so that  $\Delta\Psi/\Psi$  would be proportional to  $\Delta s/s$  (formula (6) represents in fact a similar type of relationship since it equivalently states that  $\Delta v_{i_1}/v_{i_1}$  is proportional to  $\Delta v_{i_2}/v_{i_2}$  along indifference curves). By integration one obtains that  $\Psi$  would be a power function of  $s$  (note that the function (5) is a product of power functions). Empirical evidence in many areas of sensory perception prompted Stevens (1957) to postulate the so-called power law as a general psycho-physical law, a result which is still widely accepted in the research community.

**Note.** In our definition of the relative importance of the criteria we ignored the possibility that the substitution rate between relative gains and losses may be different from the substitution rate between relative losses and gains at any point where a move along the indifference curve begins. Inequality of these rates may occur if the decision maker is more sensitive to losses than to gains, for instance.

## 5. Scale values for relative importance

In order to identify the possible scale values for the comparative judgements 'somewhat more', 'defi-

nately more', 'much more', and 'vastly more' important, we carry out an imaginary experiment. We consider two criteria  $C_{i_1}$  and  $C_{i_2}$ , and we suppose that there are two real or imaginary alternatives  $A_j$  and  $A_k$  such that the decision maker prefers  $A_j$  over  $A_k$  under  $C_{i_1}$  and  $A_k$  over  $A_j$  under  $C_{i_2}$  with an equally strong preference. The (obviously inverse) preference intensities are estimated by  $\exp(\gamma\delta_{jk})$  and  $\exp(-\gamma\delta_{jk})$  respectively, where  $\delta_{jk}$  denotes the selected gradation of his/her comparative judgement. We assume that these intensities do not depend on the performance of the alternatives under the remaining criteria. Next, we ask the decision maker to estimate his/her preference for the two alternatives under the two criteria  $C_{i_1}$  and  $C_{i_2}$  simultaneously. Let us take  $\theta_{jk}$  to denote the selected gradation of his/her comparative judgement so that his/her preference for  $A_j$  over  $A_k$  is estimated by  $\exp(\gamma\theta_{jk})$ .

Now, taking  $\omega = \omega_{i_1 i_2}$  to stand for the relative importance (the ratio of the weights) of  $C_{i_1}$  and  $C_{i_2}$ , we obtain by the geometric-mean aggregation rule that

$$\exp\left(\gamma \frac{\omega}{\omega + 1} \delta_{jk}\right) \cdot \exp\left(-\gamma \frac{1}{\omega + 1} \delta_{jk}\right) = \exp(\gamma\theta_{jk}),$$

whence

$$\omega = \frac{\delta_{jk} + \theta_{jk}}{\delta_{jk} - \theta_{jk}}.$$

Note that  $\omega$  does not depend on the scale parameter  $\gamma$ . It will be clear that  $|\theta_{jk}| < |\delta_{jk}|$  because the preference for  $A_j$  over  $A_k$  under the two criteria simultaneously cannot be greater than the preference for  $A_j$  over  $A_k$  under one of the criteria individually. It is easy to verify that  $\omega$  varies roughly between  $\frac{1}{16}$  and 16 when  $\delta_{jk}$  varies between  $-8$  and  $8$ , and  $\theta_{jk}$  between the values  $-|\delta_{jk}|$  and  $|\delta_{jk}|$ . The rather extreme value  $\omega = 15$  is obtained when  $\delta_{jk} = 8$  and  $\theta_{jk} = 7$ , which means that the decision maker's very strong preference for  $A_j$  over  $A_k$  under criterion  $C_{i_1}$  almost completely wipes out his/her equally strong but inverse preference under criterion  $C_{i_2}$ . For simplicity, we take a ratio of 16:1 to stand for 'vastly more' importance. A simple geometric sequence of values between 1 and 16, with 5 echelons, is given by the sequence 1, 2, 4, 8, 16. Hence, assuming that geometric progression is also plausible for the grada-

tions of relative importance, we take these echelons to designate 'equal', 'somewhat more', 'definitely more', 'much more', and 'vastly more' importance. If we allow for hesitations between two adjacent gradations of relative importance we eventually have a geometric sequence of scale values with progression factor  $\sqrt{2}$ . The inverse echelons are, of course, taken to designate 'equal', 'somewhat less', ..., 'vastly less' importance.

It will be obvious that the criteria can be evaluated in the same way, via pairwise comparisons, as the alternatives. We simply follow the procedure of Section 2. In the basic experiment we present two criteria  $C_{i_1}$  and  $C_{i_2}$  to decision maker  $d$ , and we ask him/her to express his/her graded comparative judgement. This is subsequently converted into the numerical value

$$r_{i_1 i_2 d} = \exp(\gamma \delta_{i_1 i_2 d}),$$

where the integer-valued gradation index  $\delta_{i_1 i_2 d}$  designates the gradations of his/her judgement as follows:

- 8:  $C_{i_1}$  vastly less important than  $C_{i_2}$ .
- 6:  $C_{i_1}$  much less important than  $C_{i_2}$ .
- ...
- + 8:  $C_{i_1}$  vastly more important than  $C_{i_2}$ .

The scale parameter  $\gamma$  (the logarithm of the progression factor) is set to  $\ln \sqrt{2}$ . It is only here that the evaluation of the criteria deviates from the evaluation of the alternatives where the scale parameter is usually set to  $\ln 2$ . Finally, we calculate the normalized criterion weights by solving a logarithmic regression problem of the type (1).

The calculation of the criterion weights in SMART is even simpler than in the Multiplicative AHP (see also Lootsma, 1993). We ask the decision maker to express the importance of the criteria in marks or grades on the scale 4, ..., 10 (grades lower than 4 are possible but they practically eliminate the corresponding criteria). Taking  $m_i$  to stand for the mark assigned to criterion,  $C_i$  we obtain that the ratio of the weights of  $C_{i_1}$  and  $C_{i_2}$  is estimated by

$$(\sqrt{2})^{m_{i_1} - m_{i_2}},$$

so that an unnormalized weight of  $C_i$  is given by  $(\sqrt{2})^{m_i}$ .

Normalization yields the desired criterion weights. We leave it to the reader to verify that the set of marks 8, 7, 4, 6 assigned to the respective criteria  $C_1, \dots, C_4$  finally yields the normalized weights 0.407, 0.287, 0.102, 0.203.

The weights calculated according to the Multiplicative AHP can immediately be used in SMART, and vice versa. This feature has been implemented and tested in the REMBRANDT program (L. Rog, Delft University of Technology) for Ratio Estimation in Magnitudes or deci-Bells to Rank Alternatives which are Non-DominatedTed.

## 6. Pre-existing preferences and values?

The research in the field of multi-criteria decision analysis follows a number of distinct approaches in order to support the decision makers in their attempts to rank the alternatives and/or to identify the preferred alternative. The descriptive approach merely tells how decision makers actually behave when they are confronted with the choice between conflicting alternatives and how multi-criteria analysis contributes to the decision process; many studies are concerned with individual and collective decision making, with an analysis of the rationality of the decision makers in various cultural contexts, and with the identification of hidden objectives. The normative approach tells how decision makers should behave and how methods for multi-criteria analysis should work, on the basis of certain fundamental axioms (transitivity, independence, etc.) and logical rules. Finally, the clinical or prescriptive approach tells how decision makers could improve the decision process and how methods for multi-criteria analysis could support such a process; key elements are the modelling of human judgement, the identification of preference intensities, and the design of decision-support systems. This short summary tends to be an oversimplification of the basic philosophies underlying the respective approaches. For more details we refer the reader to Von Winterfeldt and Edwards (1986), Bell et al. (1988), and French (1988).

The normative and the clinical approach start from the (more or less tacit) assumption that there is



a coherent, ordered system of preferences or values in the mind of the decision makers (sometimes deeply hidden) and that the methods for multi-criteria analysis fathom the system in order to generate recommendations which are compatible with it. Of course, the preferences and values of unexperienced decision makers may be unstable and volatile, multi-criteria analysis may have a disproportionate influence on the decision process, but there are certain limits to what the analysis may suggest. The idea that a coherent, well-structured system of preferences and values possibly exists in the decision maker's mind before the start of the decision process is firmly rejected by those who follow the constructivistic approach of the so-called French school in multi-criteria analysis (see Roy and Bouyssou, 1993, and Mousseau, 1993). The members of the French school refer to the normative and the clinical approach as the American school. They advocate that, in an actual decision problem, the decision maker and an analyst should jointly construct a model to shape the decision maker's preferences and values which are initially unstable and volatile or even non-existent. This implies that the results of the analysis may be highly dependent on the analyst and on his/her method. The series of ELECTRE systems has particularly been designed for the purposes of constructivism, whereas the AHP and SMART are typically tools from the American school (following the clinical approach, at least in our opinion).

An extensive discussion of these fundamental viewpoints (which, incidentally, may be due to cultural factors, see Hofstede, 1994) is beyond the scope of the present paper. We only remark here that the result of this paper does not constitute a contribution to the constructivistic approach. We started from the observation that the decision makers frequently express the relative importance of the criteria, within a coherent field of decision problems but in isolation from a particular problem in the field, and we presented a model to process the pre-existing information supplied by those decision makers. It should be noted that the treatment of the criterion weights in ELECTRE and the subsequent aggregation procedure are not always transparent (Stewart, 1992); this may be due to the constructivistic approach which ignores the possible impact of psycho-physical research (Roy, private communication).

## 7. Final remarks

The result of the present paper is that information about the relative importance of the criteria can be elicited in a distributed mode, in the absence of immediate context: when we operate in a particular field of decision problems we do not have to know the alternatives of an actual decision problem in order to judge the criteria which are usually employed in the field. This is true for the Multiplicative AHP and for SMART. They use the same criterion weights in the geometric-mean aggregation rule (Multiplicative AHP) and in the arithmetic-mean aggregation rule (SMART).

Let us conclude the paper with a critical comment. The original AHP has been designed to structure a decision problem as a hierarchy: at the top level there is the global goal, at the next level there are performance criteria to judge how well the alternatives contribute to the global goal, at the level thereafter there are sub-criteria to refine the formulation of the criteria, etc. until one arrives at the bottom level where the actual alternatives are positioned. In the present paper we have seen that the relative importance of the criteria seems to be a pre-existing concept, and we derived the scale values for the gradations of relative importance from an imaginary experiment, the pairwise comparison of two alternatives under two criteria simultaneously (Section 5). Hence, we have a basis for the analysis of a hierarchy with two levels, one for the criteria and one for the alternatives. We may reasonably assume that the relative importance of the sub-criteria, the sub-sub-criteria, etc. is also a pre-existing concept, but the question of how to scale the gradations of relative importance at these levels of the hierarchy is still open. Moreover, the weights assigned to the criteria, sub-criteria, etc. depend strongly on the structure of the hierarchy, whereas splitting a criterion appears to be an effective way to attract more weight for it (Borcherding et al., 1994). Hence, there are still many research problems to be solved before we may claim to have a basis for the analysis of hierarchies with more than two levels.

Numerical experiments involving the calculation of criterion weights on the basis of the above ideas may be found in the author's comparative study (Lootsma, 1994) which also presents a procedure to

cross-validate some current methods for multi-criteria analysis.

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